

23ECE285 DIGITAL ELECTRONICS LABORATORY L-T-P-C: 0-0-3-1

Course Objectives

- To provide hands-on experience in realizing simple logic expressions
- To demonstrate the power of logic function optimization
- To enable the implementation of combinational and sequential circuits.

Course Outcomes

CO1: Use datasheets & simulation tools effectively

CO2: Realize simple logic circuits

CO3: Design & implement combinational circuits

CO4: Design & implement sequential circuits

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3				3				2	2		2		2
CO2	3								2	2				2
CO3	3	3	2						2	2		2		2
CO4	3	3	2						2	2		2		2

Syllabus

1. Logic Gates: Implement logic gates using NAND / NOR.
2. Boolean functions: using logic gates
3. Combinational circuits: Mux, De-Mux, Code Converters, Adders & Subtractors
4. Sequential Circuits: Flip-Flops, Counters, State Machines

Experiments

1. Verification of Basic Logic Gates.
2. Realization of Basic Gates using Universal Logic Gates.
3. Simplification and Realization of a given Boolean Expression i) Using basic gates ii) SOP Using NAND gates only iii) SOP Using NOR gates only iv) POS Using NAND gates only v) POS Using NOR gates only and vi) Compare and analyze the above implementations

4. Design and verification of Adders and Subtractors.
5. Design and verification of Parallel Adder / Subtractor.
6. Design and verification of Binary to Gray code converter and vice versa.
7. Design and verification of BCD to Excess-3 code converter and vice versa.
8. Design and verification of 2-bit Magnitude Comparator.
9. Design and verification of Multiplexers
10. Implementation and verification of Half adder, full adder, half subtractor and full subtractor using multiplexers.
11. Design and verification of Flip-flops (D, T and JK flipflop).
12. Design and verification of shift Registers.
13. Design and verification of Ring and Johnson Counters.
14. Design and verification of 4-bit asynchronous Up and Down Counters

Text Book(s)

Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital logic with Verilog Design", Tata McGraw Hill Publishing Company Limited, Special Indian Edition, 2007.

Reference(s)

John F. Wakerly, "Digital Design Principles and Practices", Pearson Education, Fourth Edition, 2008.

K A Navas, "Electronic Lab Manual" – Volume 1, Fifth Edition, Prentice Hall of India, 2015.

M Morris Mano and Michael D Ciletti, "Digital Design with Introduction to the Verilog HDL", Pearson Education, Fifth Edition, Fifth Edition, 2015

Evaluation Pattern: 80:20

Assessment	Internal	External
Mid Sem Examination	20	
*Continuous Assessment Laboratory (CAL)	60	
End Semester		20

*CAL includes Lab Evaluations, Assignments